

10_lambda_functions

May 11, 2026

1 Lambda Functions

```
[1]: # user defined function
def add(a, b):
    return a + b
```

```
[2]: ?add
```

```
Signature: add(a, b)
Docstring: <no docstring>
File:
c:\users\sunny\appdata\local\temp\ipykernel_18016\630407924.py
Type:      function
```

```
[9]: add(89, 56)
```

```
[9]: 145
```

1.1 Lambda Function for addition

```
[7]: # Lambda function
# syntax: lambda arguments : expression
add_lambda = lambda a, b : a + b
```

```
[8]: add_lambda(89, 37)
```

```
[8]: 126
```

1.1.1 For Subtraction

```
[10]: subtract_lambda = lambda a, b, c: a - b - c
subtract_lambda(89, 37, 12)
```

```
[10]: 40
```

1.1.2 For Multiply

```
[12]: square = lambda x: x * x  
square(4)
```

```
[12]: 16
```

```
[13]: multiply_lambda = lambda a, b: a * b  
multiply_lambda(89, 37)
```

```
[13]: 3293
```

1.1.3 Even or Odd check

```
[14]: check = lambda x: "Even" if x % 2 == 0 else "Odd"  
check(7)
```

```
[14]: 'Odd'
```

1.1.4 For division

```
[16]: divide_lambda = lambda a, b: a / b  
divide_lambda(89, 37)
```

```
[16]: 2.4054054054054053
```

1.1.5 List sorting

```
[17]: marks = [50, 30, 90, 70]  
marks.sort(key=lambda x: x)  
marks
```

```
[17]: [30, 50, 70, 90]
```

```
[23]: number = [5, 20, 15, 18, 30, 35, 45, 56]  
# lambda function  
squared_numbers = list(map(lambda x: x**2, number))
```

```
[24]: squared_numbers
```

```
[24]: [25, 400, 225, 324, 900, 1225, 2025, 3136]
```

```
[25]: number = [5, 20, 15, 18, 30, 35, 45, 56]  
# lambda function  
cube_numbers = list(map(lambda x: x**3, number))
```

```
[26]: cube_numbers
```

```
[26]: [125, 8000, 3375, 5832, 27000, 42875, 91125, 175616]
```